

Intended Learning Objectives

In this lab you will:

- Design and simulate low-voltage and simple current mirrors.
- Compare low-voltage and simple current mirrors.
- Learn how to use hierarchical design.

PART 1: Sizing Chart

1. We can show that

$$g_m = \frac{2I_D}{V_{ov}}$$

which is based on the square-law. For a real MOSFET, if we compute V_{ov} and $\frac{2I_D}{g_m}$ they will not be equal. Let's define a new parameter called V-star (V^*) which is calculated from actual simulation data using the formula

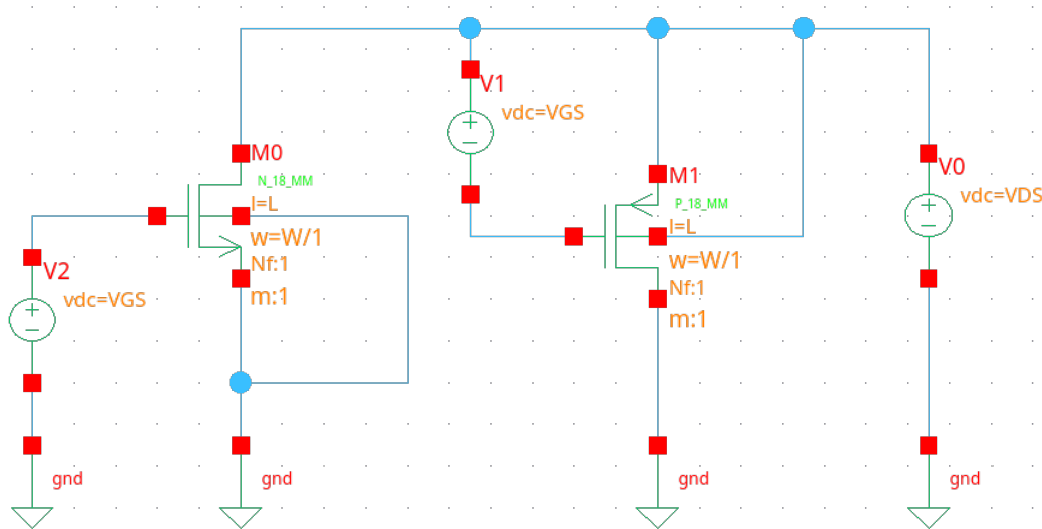
$$V^* = \frac{2I_D}{g_m} \leftrightarrow g_m = \frac{2I_D}{V^*}$$

For a square-law device, $V^*=V_{ov}$, however, for a real MOSFET they are not equal. An often used sweet-spot that provides good compromise between different trade-offs is $V^*=200\text{mV}$.

2. We want to design current mirrors with the parameters below.

Parameter	Value
L	1 μm
V^*	200 mV
Supply voltage	1.8 V
Reference current	20 μA

3. The remaining variable in the design is to calculate W. Since the square-law is not accurate, we cannot use it to determine the sizing. Instead, we will use a sizing chart generated from simulation. Create a testbench for NMOS transistor as shown below (we will use both NMOS and PMOS in this lab). Use $W = 10\mu\text{m}$ (we will understand why shortly).



4. Sweep V_{GS} from 0 to $\approx V_{TH} + 0.4V$ with 10mV step. Set $V_{DS} = V_{DD}/2$.
5. We want to compare $V^* = 2I_D/g_m$ and $V_{ov} = V_{GS} - V_{TH}$ by plotting them overlaid. Use the calculator to create expressions for V^* and V_{ov} . You can save the expressions to reuse them later.
6. Plot V^* and V_{ov} overlaid vs V_{GS} . Make sure the y-axis of both curves has the same range. You will notice that at the beginning of the strong inversion region, V^* and V_{ov} are relatively close to each other (i.e., square-law is relatively valid). For deep strong inversion (large V_{ov} : velocity saturation and mobility degradation) or weak inversion (near-threshold and sub-threshold operation) the behavior is quite far from the square-law.
7. An often used sweet-spot that provides good compromise between different trade-offs is $V^* = 200mV$. On the V^* and V_{ov} chart locate the point at which $V^* = 200mV$. Find the corresponding V_{ovQ} and V_{GSQ} .
8. Plot I_D , g_m , and g_{ds} vs V_{GS} . Find their values at V_{GSQ} . Let's name these values I_{DX} , g_{mX} , and g_{dsX} .
9. Now back to the assumption that we made that $W = 10\mu m$. This is not the actual value that we will use for our design. But the good news is that I_D is always proportional to W irrespective of the operating region and the model of the MOSFET (regardless square-law is valid or no). Thus, we can use ratio and proportion (cross-multiplication) to determine the correct width at which the current will be I_{DQ} as given in the specs. Calculate W as shown below.

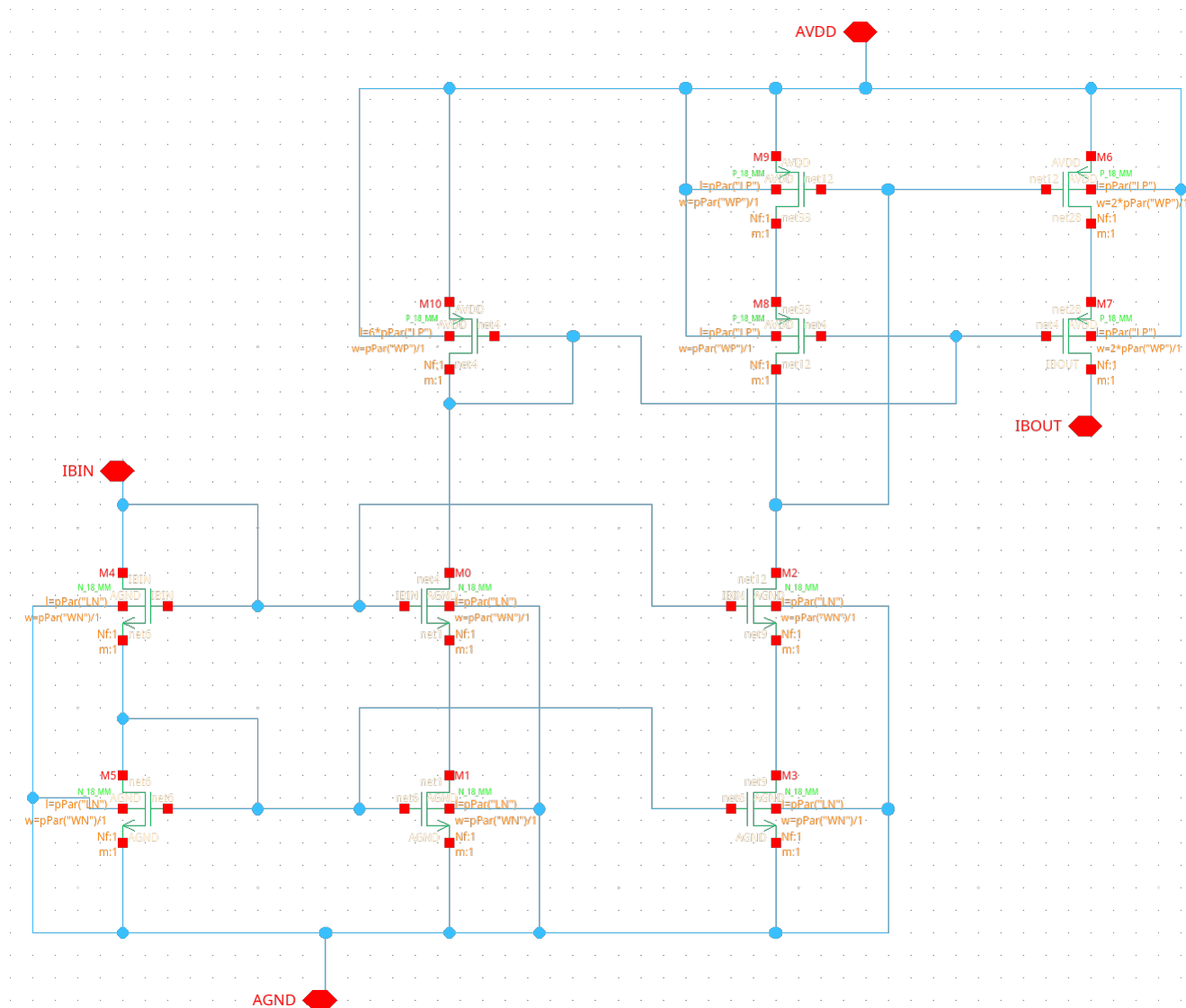
W	I_D
$10\mu m$	$I_{DX}@V_Q^*$ (from the chart)
?	$I_{DQ}=20\mu A$ (from the specs)

10. Now we are almost done with the design of the amplifier. Note that g_m is also proportional to W as long as V_{ov} is constant. On the other hand, $r_o = 1/g_{ds}$ is inversely proportional to W (I_D) as long as L is constant. Before leaving this part, calculate g_{mQ} and g_{dsQ} using ratio and proportion (cross-multiplication).

PART 2: Current Mirror

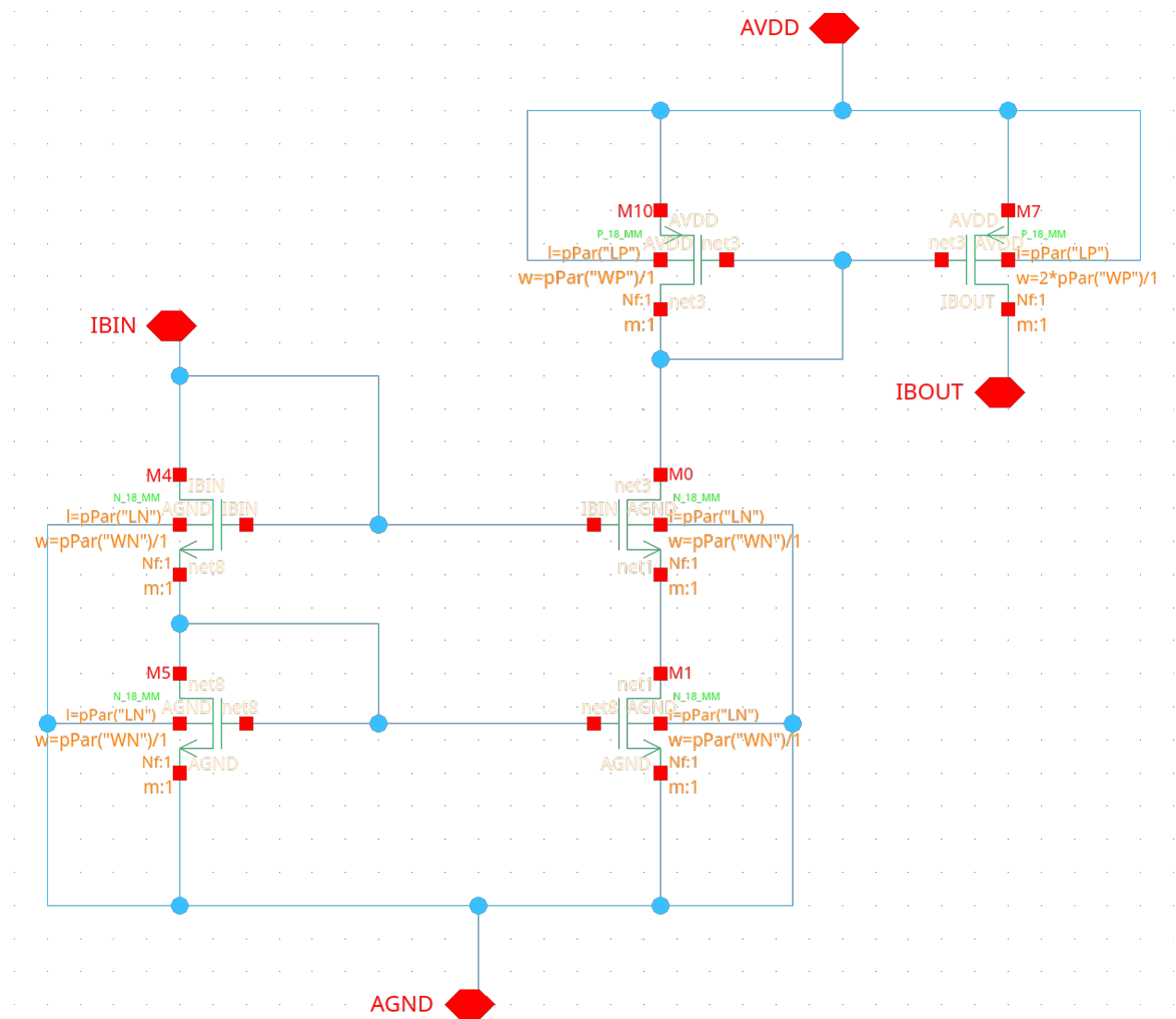
1. Create a new schematic and name it "lab_05_ws_cm". Construct the low compliance current mirror circuit shown below.
2. Construct the circuit shown below. Add labels (IBIN, IBOUT, AGND, AVDD) to nets (wires) using the hotkey "I", then create ports using the hotkey "p" or using the toolbar.

► **Cadence Hint:** For W and L we use `pPar("VariableName")`, which passes the variable to the upper level of hierarchy. The variables will appear when you instantiate the cell as explained below.

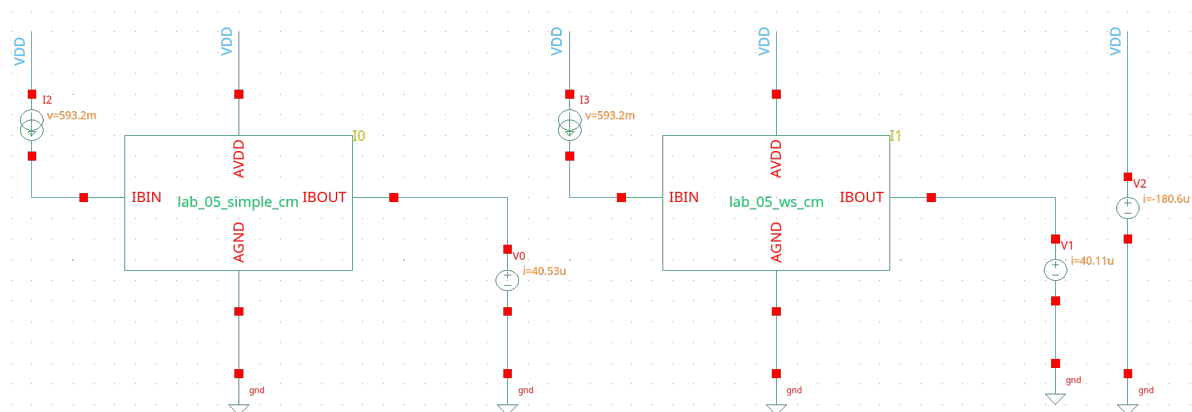


3. For the wide-swing bias transistor (the magic battery) note that we use longer length (6 * L) as shown below.
4. The current mirror takes input current IBIN and generates output current IBOUT = 2*IBIN (note the sizing of transistors in IBOUT branch).
5. Create a symbol for the CM (From the menu bar: Add -> Generate Symbol)
6. Create a testbench in a higher level of hierarchy as shown below to test the current mirror.

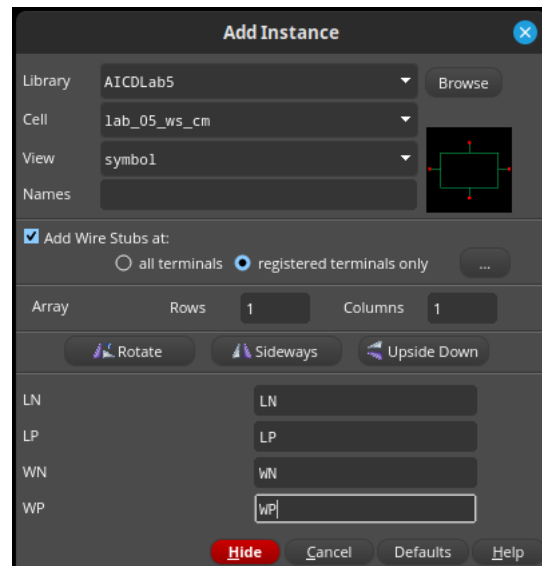
7. In the library manager copy the “lab_05_ws_cm” cell and name the new cell “lab_05_simple_cm”. Edit the new cell to design another simple current mirror using two PMOS transistors only as shown below. Create a symbol for the simple current mirror. We will compare this simple mirror with the wide swing cascode you have just designed.



8. Create a new cell for the testbench schematic “lab_05_cm_tb” to test the two current mirrors.



9. When you instantiate the cells, you can set the pPar parameters as shown below.



10. Use $I_B = 20\mu\text{A}$ and set the sizing as in Part 1.

Report the following:

1. Schematic of the two CMs with DC node voltages clearly annotated at $V_{OUT} = V_{DD}/2$ (double click the symbol to go to lower level of hierarchy and show the schematic).
2. Snapshots showing the following parameters for all transistors at $V_{OUT} = V_{DD}/2$. You may summarize the results in a table.

Parameter	Wide Swing Current Mirror		Simple Current Mirror	
	NMOS	PMOS	NMOS	PMOS
I_D				
V_{GS}				
V_{DS}				
V_{TH}				
V_{DSAT}				
g_m				
g_{ds}				
g_{mb}				
Region				

3. Check that all transistors have V_{GS} , g_m and g_{ds} as designed in Part 1.
4. Are all transistors operating in saturation?
5. Perform DC sweep (not parametric sweep) using $V_{OUT} = 0:10\text{m}:V_{DD}$. Report I_{BOUT} vs V_{OUT} for the two CMs overlaid in the same plot.

- Comment on the difference between the two circuits.
 - I_{BOUT} of the simple CM is exactly equal to $I_{BIN} \cdot 2$ at a specific value of V_{OUT} . Why?
6. Percent of error in I_{BOUT} vs V_{OUT} (ideal I_{BOUT} should be $I_{BIN} \cdot 2$) for the two CMs in the current mirror operating region ($V_{OUT} = 0$ to $V_{DD} - V^*$) overlaid in the same plot.
- Comment on the difference between the two circuits.
7. R_{out} vs V_{OUT} (take the inverse of the derivative of I_{BOUT} plot) for the two CMs in the current mirror operating region ($V_{OUT} = 0$ to $V_{DD} - V^*$) overlaid in the same plot. Use log scale on the y-axis. Add a cursor at $V_{OUT} = V_{DD}/2$.
- Comment on the difference between the two circuits.
 - Does R_{out} change with V_{OUT} ? Why?
8. Analytically calculate R_{out} of both circuits at $V_{OUT} = V_{DD}/2$. Compare with simulation results in a table.

Lab Summary

- In Part 1 you learned
 - How to generate and use design charts for NMOS and PMOS transistors.
 - How to design a PMOS common-drain amplifier.
 - How the overdrive voltage of a MOS transistor deviates from the square law in different regions of operation.
- In Part 2 you learned
 - How to do AC, DC and transient simulations of a CD amplifier.
 - How the peaking in the frequency response of a CD amplifier changes with the load capacitor and source resistance.
 - How the ringing in the transient response of a CD amplifier changes with the load capacitor and source resistance.
 - How the output impedance of a CD amplifier shows inductive behavior.